$$\frac{dy}{dx} = \frac{-(3xy + y^2)}{x^2 + xy} \qquad (1)$$

Put $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$ in (1)

$$v + x\frac{dv}{dx} = \frac{-(3x\ vx + v^2x^2)}{x^2 + x\ vx}$$

$$= \frac{-(3v + v^2)}{1 + v}$$

Now, $$x\frac{dv}{dx} = \frac{-3v - v^2}{1 + v} - v$$

$$= \frac{-3v - v^2 - v - v^2}{1 + v}$$

$$x\frac{dv}{dx} = \frac{-4v - 2v^2}{1 + v}$$

$$\frac{1 + v}{4v + 2v^2}\,dv = \frac{-dx}{x}$$

On Integration

$$\int \frac{1 + v}{4v + 2v^2}\,dv = -\int \frac{dx}{x}$$

Now, multiply 4 on both sides

$$\int \frac{4 + 4v}{4v + 2v^2}\,dv = -4\int \frac{dx}{x}$$

$$\log(4v + 2v^2) = -4\log x + \log c$$

$$4v + 2v^2 = \frac{c}{x^4}$$

$$x^4(4v + 2v^2) = c$$

Replace $v = \frac{y}{x}$

$$x^4\left(4\frac{y}{x} + 2\frac{y^2}{x^2}\right) = c$$

$$x^4\left[\frac{4xy + 2y^2}{x^2}\right] = c$$

$$c = 2x^2(2xy + y^2) \qquad (2)$$

Cost of producing a pair of shoes = ₹12

(*i.e*) $y = 12$ when $x = 2$

$$c = 8\left[48 + 144\right] = 1536$$

∴ The cost function is $x^2(2xy + y^2) = 768$

Example 4.19

The marginal revenue 'y' of output 'q' is given by the equation $\frac{dy}{dq} = \frac{q^2 + 3y^2}{2qy}$. Find the total Revenue function when output is 1 unit and Revenue is ₹5.

Solution:

Given that $MR = \frac{dy}{dq} = \frac{q^2 + 3y^2}{2qy}$ (1)

Put $y = vq$ and $\frac{dy}{dq} = v + q\frac{dv}{dq}$ in (1)

Now (1) becomes

$$v + q\frac{dv}{dq} = \frac{q^2 + 3v^2q^2}{2q\ vq}$$

$$= \frac{1 + 3v^2}{2v}$$

$$q\frac{dv}{dq} = \frac{1 + 3v^2}{2v} - v$$

$$= \frac{1 + 3v^2 - 2v^2}{2v}$$

$$= \frac{1 + v^2}{2v}$$

$$\frac{2v}{1 + v^2}\,dv = \frac{dq}{q}$$

On Integration

$$\int \frac{2v}{1 + v^2}\,dv = \int \frac{dq}{q}$$

$$\log(1 + v^2) = \log q + \log c$$

$$1 + v^2 = cq$$

Replace $v = \frac{y}{q}$

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